

AI-Based Anomaly Detection for Predictive HACCP Using IoT-Based Environmental Monitoring

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MSc Artificial Intelligence

Module Code

CMI3420

Module Title

Individual Project

University of Huddersfield

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1. Project Definition

1.1 Context and Rationale

Food safety is still a big issue around the world. The World Health Organization (2024) says roughly 600 million people end up sick from eating contaminated food each year. Even though factories follow strict hygiene rules and use HACCP systems, things like *Listeria monocytogenes* still manage to cause problems. In a lot of places, they're still doing hygiene checks the old-fashioned way taking swabs and sending them off to a lab. The trouble is, you're looking at 24 to 72 hours before you get the results back. By then, the food might have already left the factory and ended up on shop shelves.

That's where newer tech like AI and the Internet of Things (IoT) comes in. They offer a way to stay ahead of problems instead of just reacting after the fact. Temperature, humidity, even how often someone opens a door all of that affects bacteria growth and contamination risk. Research by Kanicheril Ambikalekshmi et al. (2025) and Mu et al. (2024) found that machine learning can go through environmental data and spot unusual patterns long before a routine inspection would catch anything.

Although several studies have looked at using AI and IoT for food safety, very few have tested these systems using real industrial swab records from working food production sites. This project aims to fill that gap by combining environmental monitoring data with historical swab records to see whether AI-based anomaly detection can help spot hygiene risks earlier in real manufacturing settings.

1.2 High-Level Aim

The aim of this project is to develop and evaluate an AI-based anomaly detection system that combines environmental monitoring data with historical swab records to support predictive HACCP monitoring in food manufacturing environments.

1.3 Scope

This project focuses on the development and evaluation of an AI-supported anomaly detection approach for food hygiene monitoring. The work includes analysing historical environmental data together with industrial swab records and comparing machine learning models for anomaly detection.

Three machine learning approaches Random Forest, LSTM, and Isolation Forest will be evaluated using precision, recall, and F1-score. The project will investigate whether environmental patterns appear before abnormal swab outcomes and assess the practical feasibility of predictive monitoring in food manufacturing environments.

The study also includes a small-scale IoT prototype using DHT22 temperature and humidity sensors together with a reed switch door sensor to demonstrate how real-time environmental monitoring could support future predictive HACCP systems. The project does not include deployment within a live production environment or full industrial real-time alerting infrastructure beyond the prototype stage.

2. Review and Methodology

2.1 Knowledge Review Strategy

As part of the literature review for the research, I am going to employ both Google Scholar and Scopus as primary sources. They are conveniently accessible via the university login page. Specifically, the literature review will concentrate on four topics: functioning and limitations of HACCP frameworks, application of AI and machine learning algorithms in food safety, Internet of Things-based environmental monitoring, and various methods of detecting anomalies in the data from the time series.

Key keywords used for the searches will include "AI anomaly detection food safety", "LSTM environmental monitoring", "IoT HACCP", and "predictive contamination detection". For the most part, the research will rely on studies from 2021 to 2026 inclusive, but some foundational sources from previous periods may also be included where necessary. Overall, I aim to examine no less than 20 peer-reviewed works in connection with the methodology adopted within the dissertation.

2.2 Investigation and Data Analysis

This project follows a quantitative experimental design combining continuous IoT environmental readings with historical swab records to train and evaluate machine learning models.

Data Sources

The primary dataset is historical swab records from Hygiene Improvement Solutions, subject to ethical approval and a signed data sharing agreement. Two confirmed fallbacks are available if needed FSA open data at data.food.gov.uk and UCI environmental sensor data at archive.ics.uci.edu both requiring no additional permissions.

All datasets will be cleaned, timestamp-aligned, and brought to a minimum 90% completeness before modelling begins. Swab results will be time-aligned with the 24-hour sensor window before each sample collection to identify pre-contamination environmental signatures.

IoT Hardware

A Raspberry Pi 4 with two DHT22 sensors and one reed switch door sensor will collect readings every five minutes and transmit them to Microsoft Azure IoT Hub. A local CSV backup will be saved to the Raspberry Pi SD card continuously. If data loss exceeds 5% during any collection period, the affected timeframe will be documented and excluded from analysis.

Machine Learning Models

Random Forest and Isolation Forest are the two primary models both well established and achievable within the project timeline. LSTM will be developed as a third model where time

permits. If tuning proves too time-consuming, it will be documented as a planned extension rather than a failed deliverable. All training will be done locally in Jupyter Notebook using Python, scikit-learn, and TensorFlow. Azure ML will be considered for deployment if time allows but is not a core requirement.

All models will be evaluated using precision, recall, F1-score, and detection delay. Where only date-level swab data is available, detection delay will be measured in days and noted as a limitation. SHAP analysis will explain which variables drove each prediction.

2.3 Professional and Ethical Issues

All swab records will be fully anonymised before access. No personal employee data will be collected. A formal data sharing agreement will be signed before any transfer takes place. All data will be stored on password-protected devices and Microsoft Azure, fully compliant with UK GDPR and the Data Protection Act 2018. University ethical approval will be obtained before any company data is accessed. The signed Ethical Review Form is included in Appendix.

3. Key Deliverables

3.1 Software

The project will produce AI-based anomaly detection models, a small-scale IoT monitoring prototype, processed environmental and swab datasets, and evaluation outputs including graphs, tables, and performance visualisations.

3.2 Final Dissertation

A final dissertation of approximately 12,000–15,000 words will be submitted by 1 September 2026. The dissertation will present the methodology, findings, evaluation, and practical implications of the research.

3.3 Relationship to Course

The project integrates knowledge from several MSc modules, including Artificial Intelligence, Machine Learning, Data Mining, IoT Systems, and Cloud Computing. It combines theoretical concepts with practical implementation through data analysis, anomaly detection modelling, and environmental monitoring design.

4. Project Plan

4.1 Objectives and Milestones

The project is organised around four clear objectives with target dates and expected outcomes.

Objective 1 — Literature Review and Data Preparation

Weeks 1–12 | Milestone: 15 June 2026

Review at least 20 academic papers related to HACCP, AI anomaly detection, IoT monitoring, and food safety. Collect and prepare all datasets before model development begins, ensuring a minimum 90% data completeness. The main dataset will be historical swab records from Hygiene Improvement Solutions, subject to approval. FSA open data and UCI environmental sensor datasets will be used as fallback sources if needed.

Objective 2 — IoT Prototype Development

Weeks 12–15 | Milestone: 6 July 2026

Develop a small Raspberry Pi prototype using two DHT22 temperature and humidity sensors together with one reed switch door sensor. The system will record environmental readings every five minutes, save local backup files, and send data to Microsoft Azure IoT Hub. The target is to keep data loss below 5% during testing.

Objective 3 — AI Model Development and Comparison

Weeks 15–20 | Milestone: 10 August 2026

Develop and compare Random Forest and Isolation Forest as the two main machine learning models. LSTM will also be explored as a third model if time allows. If LSTM development becomes too time-consuming, it will be discussed as future work rather than treated as a failed task.

The models will be evaluated using precision, recall, F1-score, and detection delay. SHAP analysis will also be used to explain which environmental factors influenced the model predictions. The target is to achieve at least 85% accuracy with the primary models.

Objective 4 — Evaluation and Feasibility Assessment

Weeks 20–24 | Milestone: 25 August 2026

Evaluate the overall performance of the AI models and assess whether the proposed system could realistically support predictive HACCP monitoring in food manufacturing environments. Any limitations related to timestamp accuracy or dataset quality will be discussed in the dissertation. The final dissertation will be submitted by 1 September 2026.

4.2 Risk Mitigation Strategy

The project risks mainly relate to industrial data access, hardware reliability, technical complexity, and project timescales. Several mitigation measures have been prepared to reduce the impact of these risks and maintain project continuity.

Table 1 risk mitigation

Risk	Likelihood	Impact	Risk Level	Mitigation
Company denies or delays data access	Medium	Major	High	FSA open data and UCI datasets confirmed and ready as immediate fallback
IoT sensor failure or data loss	Low	Moderate	Medium	Spare components ordered in advance, synthetic temperature and humidity data generated from historical patterns used as substitute
Azure integration more complex than expected	Medium	Moderate	Medium	Core models built and tested locally first, Azure added as extension
ML models do not reach the targeted performance threshold	Low	Moderate	Medium	Three models compared findings remain valid as a comparative study regardless of individual accuracy levels
Time pressure from competing deadlines	Medium	Major	High	Periodical milestones set from the start; dissertation writing runs alongside development
Ethics approval delayed	Low	Major	Medium	Open datasets used immediately while approval is pending, no company data accessed until approval is granted

The full University Risk Analysis and Management form is included in Appendix A.

4.3 Detailed Timetable

A detailed project timetable and Gantt chart have been prepared to monitor progress across literature review, data preparation, prototype development, AI model training, evaluation, dissertation writing, and final submission stages.

Phase	Activity	Deliverable	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Planning	Proposal Submission	Due 3 Mar								
Literature	Literature Review									
Ethics & ToR	Ethics Form	Due 15 May								
Ethics & ToR	Terms of Reference	Due 22 May								

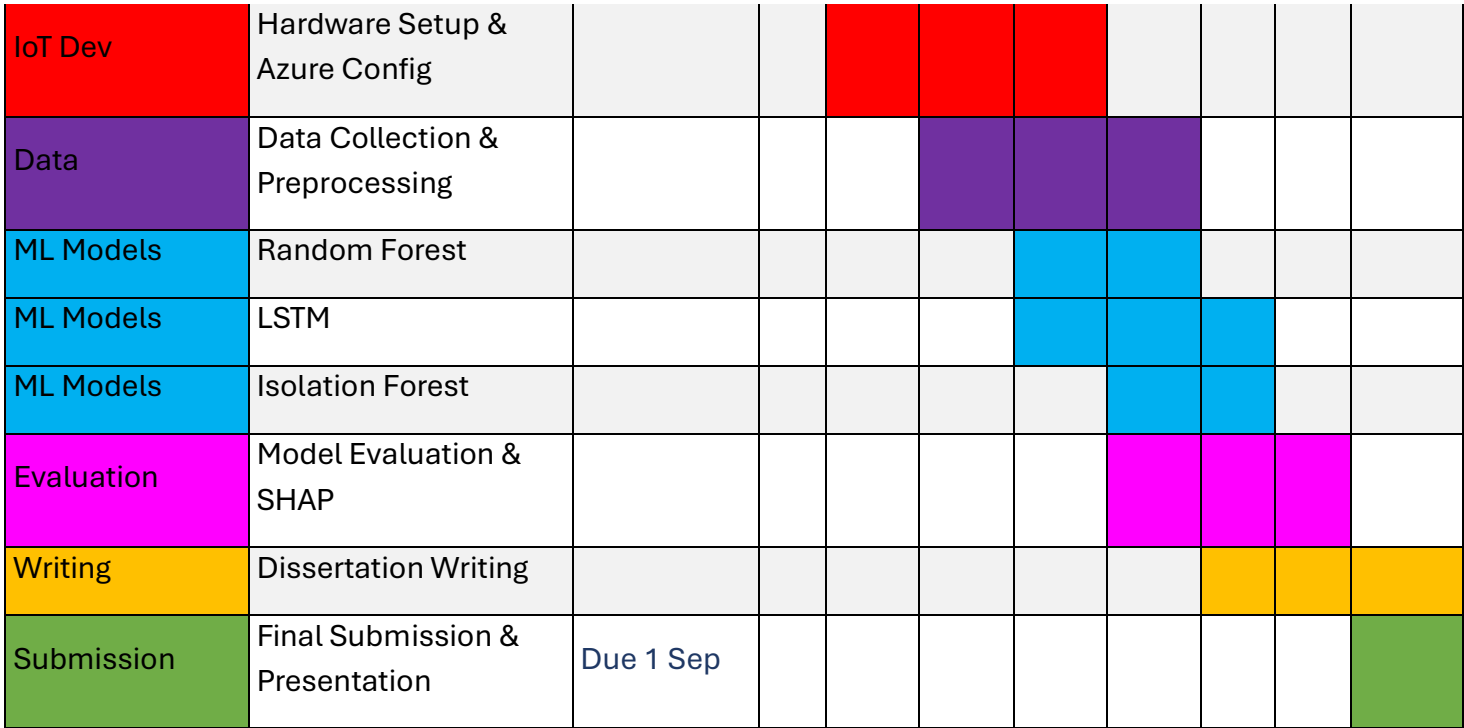


Figure 1Project Timeline and Gantt Chart (February–September 2026)

5. Resources

5.1 Documentation Standards

The dissertation will follow University of Huddersfield formatting requirements together with APA 7 referencing guidelines for all citations and references. Microsoft Word will be used for report writing and formatting, while GitHub version control will support code organisation, backup management, and project tracking throughout development.

5.2 Software, Hardware and Lab Access

For the IoT prototype, I'm using a Raspberry Pi 4, a couple of DHT22 sensors to track temperature and humidity, and a simple reed switch to tell me when the door's been opened. The whole setup should come in at under £50, which is cheap enough that it could be used in a small factory without breaking the bank.

On the software side, I'll be writing everything in Python it's what I'm most comfortable with. For building and testing the models, I'm using TensorFlow and Scikit-learn, and I'll do most of my experimenting and training in Jupyter Notebook because it's handy for trying things out step by step. I'm also using Microsoft Azure IoT Hub to collect and store the sensor data in the cloud when needed, but I'm not relying on it for anything critical.

5.3 Data Sources

The primary dataset will consist of historical swab records from Hygiene Improvement Solutions, subject to ethical approval and a signed data sharing agreement. Confirmed fallback datasets are also available, including FSA open food safety data available at data.food.gov.uk and UCI environmental sensor datasets available at archive.ics.uci.edu. These sources will support model development and validation if company data access is delayed or restricted.

6. References

Microsoft. (2025). *Azure IoT Hub documentation*. Microsoft Learn. Retrieved May 2026, from <https://learn.microsoft.com/en-us/azure/iot-hub/>

National Advisory Committee on Microbiological Criteria for Foods. (1997). Hazard analysis and critical control point principles and application guidelines. *Journal of Food Protection*, 61(9), 1246–1259. <https://doi.org/10.4315/0362-028X-61.9.1246>

University of California Irvine Machine Learning Repository. (2025). *Environmental sensor telemetry data set*. <https://archive.ics.uci.edu/>

UK Food Standards Agency. (2025). *Food hygiene rating scheme open data*. <https://data.food.gov.uk/>

World Health Organization. (2024). *Food safety*. <https://www.who.int/news-room/fact-sheets/detail/food-safety>

7. Appendices

2 University of Huddersfield
3 School of Computing and Engineering

1 PROJECT ETHICAL REVIEW FORM

Applicable for all research, master's and undergraduate projects

Project Title:	AI-Based Anomaly Detection for Predictive HACCP Using IoT-Based Environmental Monitoring
Student:	Asmerom Berhane
Course/Programme:	MSc Artificial Intelligence
Department:	School of Computing and Engineering
Supervisor:	Dr Ann Smith
Project Start Date:	10 February 2026

16 ETHICAL REVIEW CHECKLIST

	Yes	No
1 Are there problems with any participant's right to remain anonymous?		X
2 Could a conflict of interest arise between a collaborating partner or funding source and the potential outcomes of the research, e.g. due to the need for confidentiality?		X
3 Will financial inducements be offered?		X
4 Will deception of participants be necessary during the research?		X
5 Does the research involve experimentation on any of the following?		
(i)		
animals?		X
(ii)		
animal tissues?		X
(iii)		
human tissues (including blood, fluid, skin, cell lines)?		X
6 Does the research involve participants who may be particularly vulnerable, e.g. children or adults with severe learning disabilities?		X
7 Could the research induce psychological stress or anxiety for the participants beyond that encountered in normal life?		X
8 Is it likely that the research will put any of the following at risk:		
(i)		
living creatures?		X
(ii)		
stakeholders (disregarding health and safety, which is covered by Q9)?		X
(iii)		
the environment?		X
(iv)		
the economy?		X
9 Having completed a health and safety risk assessment form and taken all reasonable practicable steps to minimise risk from the hazards identified, are the residual risks acceptable (Please attach a risk assessment form – available at the end of this document)	X	

17 STATEMENT OF ETHICAL ISSUES AND ACTIONS

If the answer to any of the questions above is yes, or there are any other ethical issues that arise that are not covered by the checklist, then please give a summary of the ethical issues and the action that will be taken to address these in the box below. If you believe there to be no ethical issues, please enter "NONE".

STATEMENT BY THE STUDENT

I believe that the information I have given in this form on ethical issues is correct.

Signature: Asmerom Berhane Date: 22/05/2026

AFFIRMATION BY THE SUPERVISOR

I have read this Ethical Review Checklist, and I can confirm that, to the best of my understanding, the information presented by the student is correct and appropriate to allow an informed judgement on whether further ethical approval is required.

Signature: A Smith Date: 26/5/2026

SUPERVISOR RECOMMENDATION ON THE PROJECT'S ETHICAL STATUS

Having satisfied myself of the accuracy of the project ethical statement, I believe that the appropriate action is:

The project proceeds in its present form	
The project proposal needs further assessment by an Ethical Review Panel. The Supervisor will pass the form to the Ethical Review Panel Leader for consideration.	

RETENTION OF THIS FORM

- The Supervisor must retain a copy of this form until the project report/dissertation is produced.
- The student must include a copy of the form as an appendix in the report/dissertation.

OUTCOME OF THE ETHICAL REVIEW PANEL PROCESS, WHERE REQUIRED

Tick One

1.

Approved. The ethical issues have been adequately addressed and the project may commence.

2.

Approved subject to minor amendments. The required amendments are stated in the box below. The project may proceed once the form has been amended in line with the requirements and signed by the Supervisor in the box immediately below to confirm this.

I confirm, as Supervisor, that the amendments required have been made:

Signature: _____ Date: _____

3.

Resubmit. The areas requiring further action are stated in the box below. The project may not proceed until the form has been resubmitted and approved.

4.

Reject. The reasons why it will not be possible to address the ethical issues adequately are stated in the box below.

For any of the outcomes 2, 3 or 4 above, please provide a statement in the box below.

AFFIRMATION BY THE REVIEW PANEL LEADER

I approve the decision reached above by the review panel members:

Signature: _____ Date: _____

THE UNIVERSITY OF HUDDERSFIELD: RISK ANALYSIS & MANAGEMENT

ACTIVITY: AI-Based Anomaly Detection for Predictive HACCP Using IoT-Based Environmental Monitoring 1.				Name: Asmerom Berhane	
LOCATION: University of Huddersfield				Date: 22/05/26	Review Date: 22/05/2026
18.	Hazard(s) Identified	Details of Risk(s)	People at Risk	Risk management measures	Other comments

Confidential industrial data	Unauthorised access to swab records or factory information	Company stakeholders	All data anonymised and stored on password-protected	No company or employee names collected
Sensor failure or data loss	Missing environmental readings during collection	Researcher	Backup public datasets and simulated sensor data prepared as fallback	Low hardware risk
Azure/cloud integration issues	Delay in data transmission or storage	Researcher	Core models tested locally before Azure integration	Azure used only where appropriate
Project time pressure	Delays affecting project completion	Researcher	Periodic milestones and parallel writing schedule	Gantt chart included in proposal
Ethics approval delay	Delay in accessing company data	Researcher	Public FSA and UCI datasets used while approval pending	No company data accessed before approval
Electrical equipment Raspberry Pi and sensors	Minor electrical risk during hardware setup	Researcher only	Standard 5V low-voltage device with no significant hazard	Minimal physical risk
Screen and computer use over long periods	Eye strain or repetitive strain injury	Researcher only	Regular breaks and university health & safety guidance followed	Standard workstation precautions